

Relatório de Alinhamento de Sequências (Smith-Waterman)

Bioinformática

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1 Matriz de Scores (Todos contra Todos)

Abaixo apresentamos a matriz de scores resultante da comparação de todas as sequências entre si. Esta tabela exibe a pontuação máxima obtida no alinhamento local entre cada par de sequências.

Matriz de Scores Smith-Waterman (Todos contra Todos)

	ACTGACTA	TGACTGA	GACTTAC	CCTGACAA	TTGACTC	ACTTGAAC	GACTGGA	TACTGAC	GGACTTA	CTGACTG
ACTGACTA	0.0	10.0	9.0	11.0	10.0	8.0	8.0	12.0	8.0	12.0
TGACTGA	10.0	0.0	9.0	8.0	10.0	8.0	10.0	10.0	9.0	12.0
GACTTAC	9.0	9.0	0.0	7.0	8.0	8.0	8.0	9.0	12.0	8.0
CCTGACAA	11.0	8.0	7.0	0.0	8.0	7.0	6.0	10.0	6.0	10.0
TTGACTC	10.0	10.0	8.0	8.0	0.0	8.0	8.0	8.0	8.0	10.0
ACTTGAAC	8.0	8.0	8.0	7.0	8.0	0.0	9.0	8.0	8.0	6.0
GACTGGA	8.0	10.0	8.0	6.0	8.0	9.0	0.0	8.0	8.0	10.0
TACTGAC	12.0	10.0	9.0	10.0	8.0	8.0	8.0	0.0	7.0	10.0
GGACTTA	8.0	9.0	12.0	6.0	8.0	8.0	8.0	7.0	0.0	8.0
CTGACTG	12.0	12.0	8.0	10.0	10.0	6.0	10.0	10.0	8.0	0.0

Figure 1: Matriz de Scores Smith-Waterman

2 Matriz de Melhores Alinhamentos Locais

Nesta seção, visualizamos os alinhamentos literais (bases alinhadas) correspondentes aos scores máximos obtidos. Devido ao tamanho, a imagem foi ajustada para melhor visualização.

Matrizes de Alinhamentos (Melhores Alinhamentos Locais)

	ACTGACTA	TGACTGA	GACTTAC	CCTGACAA	TTGACTC	ACTTGAAC	GACTGGA	TACTGAC	GGACTTA	CTGACTG
ACTGACTA		ACTGA ACTGA	ACTGAC ACTTAC	CTGACTA CTGACAA	TGACT TGACT	AC-TGA ACTTGA	ACTG ACTG	ACTGAC ACTGAC	GACT GACT	CTGACT CTGACT
TGACTGA	TGACT TGACT		GACTGA GACTTA	TGAC TGAC	TGACT TGACT	AC-TGA ACTTGA	GACTG GACTG	ACTGA ACTGA	GACTGA GACTTA	TGACTG TGACTG
GACTTAC	ACTTAC ACTGAC	GACTTA GACTGA		CTTAC CTGAC	GACT GACT	ACTT ACTT	GACT GACT	ACTTAC ACTGAC	GACTTA GACTTA	GACT GACT
CCTGACAA	CTGACAA CTGACTA	CTGA CTGA	CTGAC CTTAC		TGAC TGAC	CCTGA CTTGA	CTG CTG	CTGAC CTGAC	GAC GAC	CTGAC CTGAC
TTGACTC	TGACT TGACT	TGACT TGACT	GACT GACT	TGAC TGAC		TTGA TTGA	GACT GACT	TGAC TGAC	GACT GACT	TGACT TGACT
ACTTGAAC	ACTTGA AC-TGA	ACTTGA AC-TGA	ACTT ACTT	CTTGA CCTGA	TTGA TTGA		ACTTGA ACTGGA	ACTTGA AC-TGA	ACTT ACTT	ACT ACT
GACTGGA	GACT GACT	GACTG GACTG	GACT GACT	GAC GAC	GACT GACT	ACTGGA ACTTGA		ACTG ACTG	GACT GACT	GACTG GACTG
TACTGAC	ACTGAC ACTGAC	ACTGA ACTGA	ACTGAC ACTTAC	CTGAC CTGAC	TGAC TGAC	AC-TGA ACTTGA	ACTG ACTG		ACTGA ACTTA	CTGAC CTGAC
GGACTTA	GACT GACT	GACTTA GACTTA	GACTTA GACTTA	GAC GAC	GACT GACT	ACTT ACTT	GACT GACT	ACTTA ACTGA		GACT GACT
CTGACTG	CTGACT CTGACT	TGACTG TGACTG	GACT GACT	CTGAC CTGAC	TGACT TGACT	TGA TGA	GACTG GACTG	CTGAC CTGAC	GACT GACT	

Figure 2: Tabela com os alinhamentos gerados

3 Top 2 Melhores Alinhamentos

Destaque para os dois melhores alinhamentos encontrados em todo o conjunto de dados (excluindo a comparação de uma sequência com ela mesma).

Top 2 Melhores Alinhamentos (Excluindo Auto-comparação)

Sequência A	Sequência B	Alinhamento	Score
ACTGACTA	TACTGAC	ACTGAC ACTGAC	12
ACTGACTA	CTGACTG	CTGACT CTGACT	12

Figure 3: Ranking dos Top 2 Alinhamentos